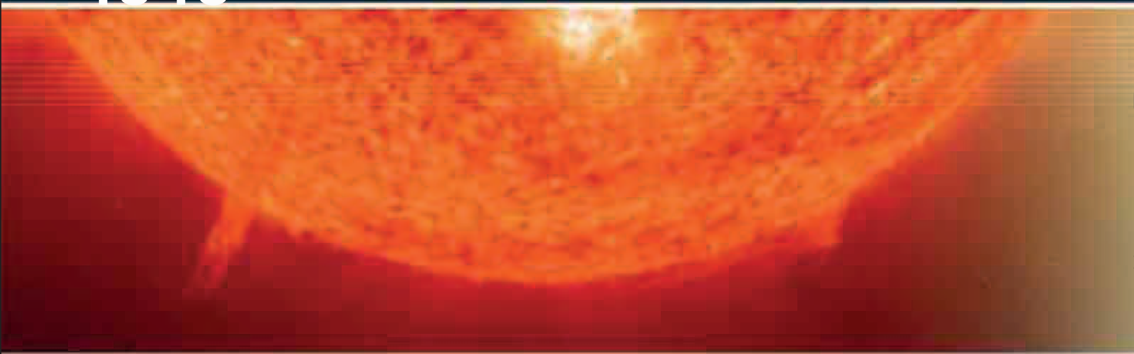


AT REST, HOWEVER, IN THE **MIDDLE OF EVERYTHING** IS **THE SUN.**

Nicolaus Copernicus, Polish astronomer, from *De Revolutionibus Orbium Coelestium,* 1543

Copernicus's idea that all the planets revolve around the Sun, rather than Earth being at the center of the Universe, was a departure from conventional astronomy and challenged the authority of the Church.

A CENTURY AFTER GUTENBERG REVOLUTIONIZED PRINTING with the invention of **movable type** (see 1450–67), scientists were able to publish their work for a mass readership, giving new ideas wider influence. The year 1543 was a milestone in **scientific publishing**, when several important books first appeared. Two books stood out—Nicolaus Copernicus's *De Revolutionibus Orbium Coelestium* (*On the Revolutions of Celestial Bodies*) and Andreas Vesalius's *De Humani Corporis Fabrica* (*On the Structure of the Human Body*). They are often seen as marking the beginning of a **new scientific age**, as they called into question the conventional authorities on **astronomy** and **anatomy**.

Up to that time, most astronomers believed that Earth was at the center of the Universe—a view put forward by **Ptolemy** in the 2nd century. **Copernicus**, however, calculated that **Earth, and all the planets, revolved around the Sun**. He had been working on this idea since about 1510, and by the 1530s had put together mathematical calculations to support his argument. However, he was reluctant to publish his theory because it **challenged convention** and went against the Church. He was persuaded to publish *De Revolutionibus* by Georg Rheticus, an Austrian mathematician who had come to study with him. It is said that Copernicus was presented with its first edition on his deathbed.

An expensive book, *De Revolutionibus* sold only a few hundred copies and did not have an immediate impact. However, Copernicus's mathematical arguments for a heliocentric (sun-centered) Universe were soon accepted by most astronomers, leading to a rift between them and the Church.

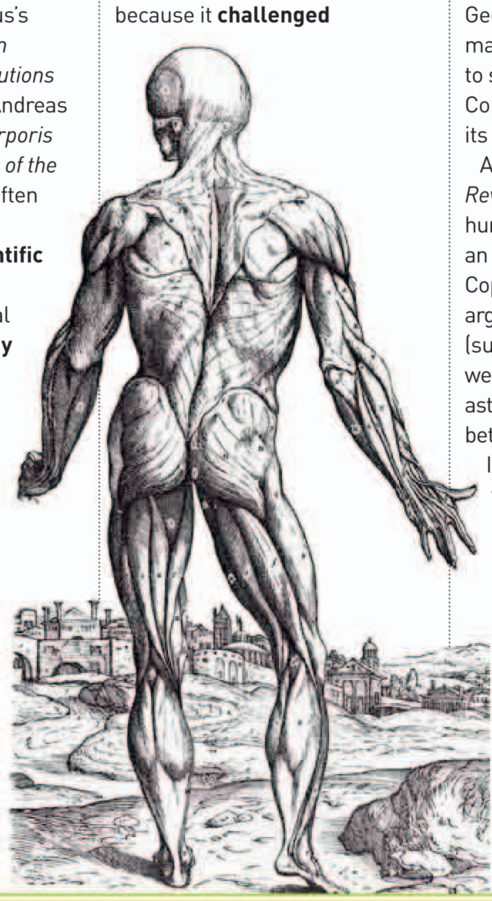
In contrast, **Vesalius** was 28 years old when he published his comprehensive seven-volume study of human anatomy, *De Humani Corporis Fabrica*. This was the **first book of human anatomy to be fully illustrated**, showing in detail what Vesalius had discovered in his dissections of human bodies. Unlike Copernicus's work, *De Humani* sold well, and Vesalius published a single-volume summary of the book later in 1543.

This year saw groundbreaking publications in the field of mathematics as well. Italian engineer and mathematician **Niccolò Fontana Tartaglia** published his translation of Euclid's *Elements* into Italian, the first translation of that work into a modern European language.

Welsh mathematician **Robert Recorde** published ***The Ground of Artes***, the first printed book on mathematics in English. It was to remain a standard textbook for more than a century.

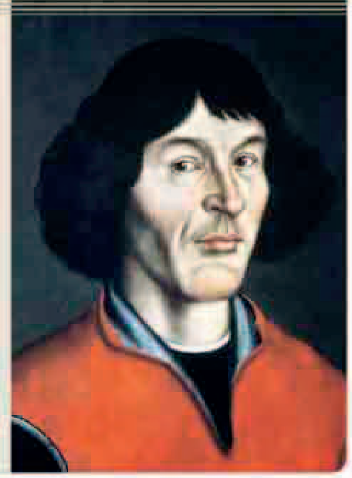
400 THE FIRST **PRINT RUN** OF COPERNICUS'S *DE REVOLUTIONIBUS ORBIUM COELESTIUM*

The heliocentric Universe
In De Revolutionibus Orbium Coelestium, *Copernicus used mathematics and astronomical observations to show how Earth and the other five planets move in circular orbits around the Sun.*



De Humani Corporis Fabrica
Vesalius's treatise on human anatomy was lavishly illustrated with detailed drawings of various stages of dissection. The figures are drawn in poses similar to the allegorical paintings of the time.

NICOLAUS COPERNICUS (1473–1543)



Born in Torun, Poland, into a German family, Nicolaus Copernicus was brought up by his uncle after his father's death. He studied law in Bologna and medicine in Padua. Copernicus lectured in mathematics in Rome before returning to Poland to work as a physician. He developed the idea of a heliocentric Universe, but had only just published his work when he died in 1543.

May Nicolaus Copernicus suggests the **Sun is at the center of the Universe**

Andreas Vesalius publishes *De Humani Corporis Fabrica*, a pioneering book on human anatomy